

# RAHUL DAS.

is a



nextleap

Product Manager Fellowship

# bookmyshow



Previously **Go for ticketing**, Founded in **2002** by **Ashish, Parikhsit** and **Rajesh**

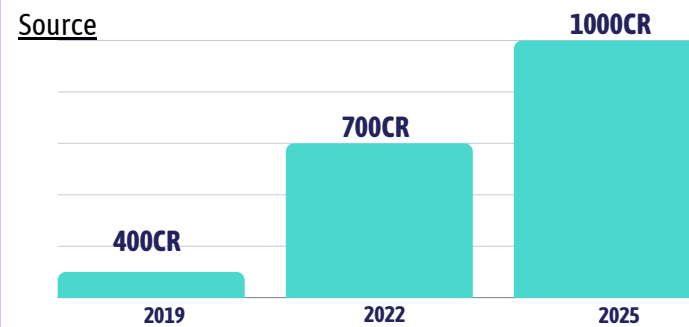
**30 million**  
users in 2024

**50 million**  
app  
downloads

**15 million**  
monthly  
ticket sales

## Live concert market size in India

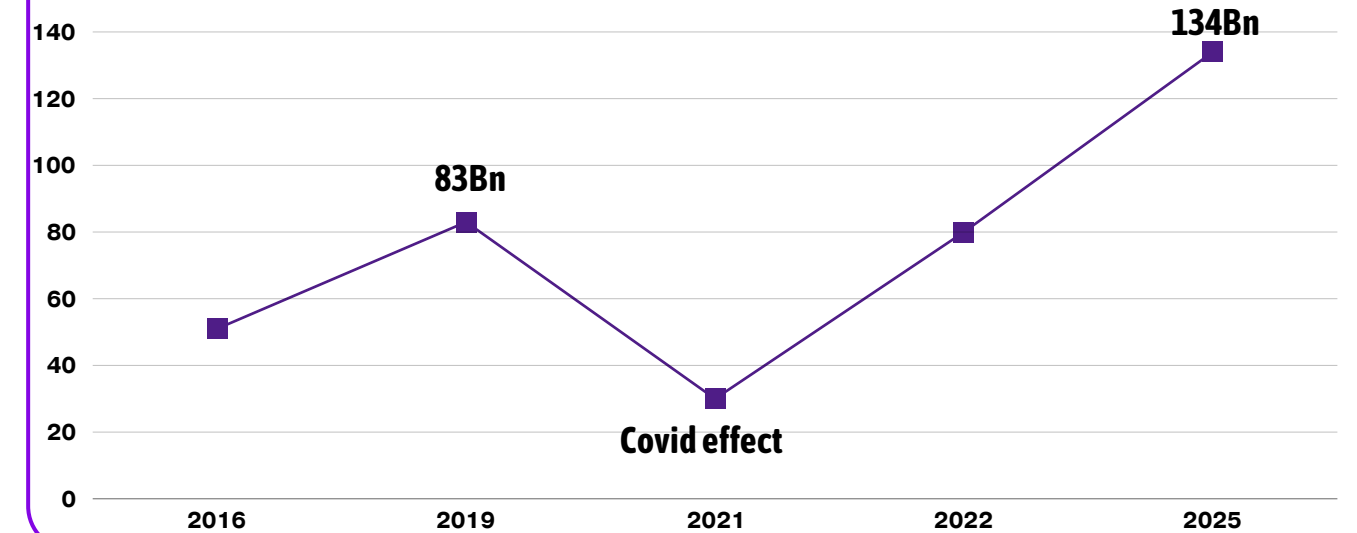
Source



Concert tourism has seen over **30%** growth in recent years

## Live events market across India

Source



BookMyShow is India's top platform for **booking movie** and **event tickets**. It offers a quick, digital ticketing experience.

## Competitors of BMS



**PVR** offers a seamless experience by **combining ticket booking** with exclusive **loyalty programs and member benefits**.

**Insider.in** specializes in live events, providing a curated selection of concerts and festivals

**TicketNew** allows users to book up to **10 tickets per transaction**, which is advantageous for group bookings.

**District** by **Zomato** is a new app designed to enhance the **"going-out"** experience by integrating dining services with ticketing for movies, sports, and live events.

## Current offerings of BMS



- Event tickets
- Live concert
- Sports events
- Standup comedy
- Theatrical performances
- Workshop and classes
- Cultural events

## Why **Zomato's district app** might be a threat for BMS.

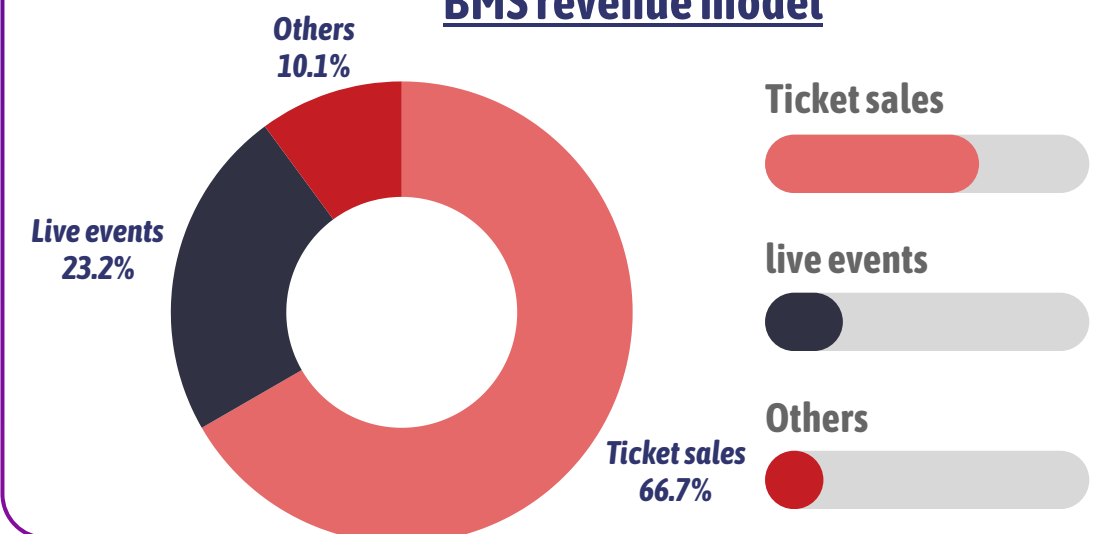
- Combines tickets, dining, and shopping**, more convenient than BookMyShow.
- Leverages its **large user base** to grow District's entertainment users.
- Recent **ticketing problems** may push users to District.
- Acquiring Paytm's ticketing** strengthens Zomato's challenge to BookMyShow.

## Business model of BMS

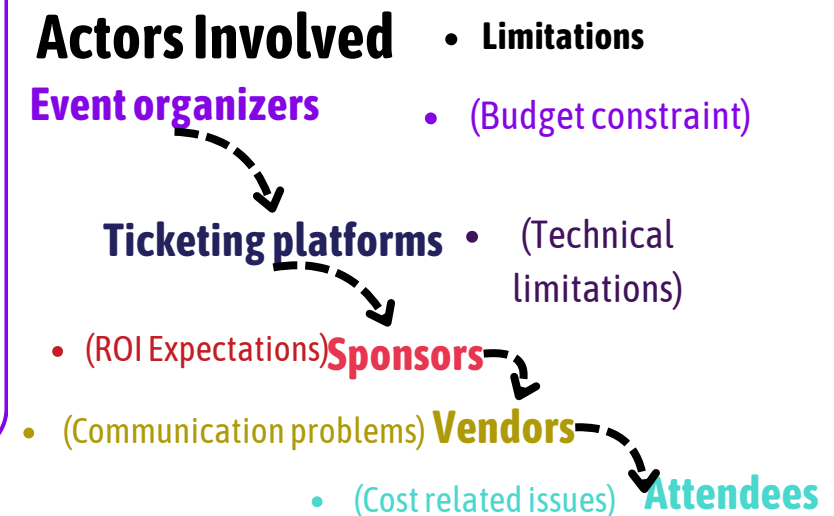
Source

- Ticket Sales:** Commissions from movie and event tickets.
- Live Events:** Revenue from organizing and promoting events.
- Others:** Income from ads, merchandise, and partnerships.

## BMS revenue model

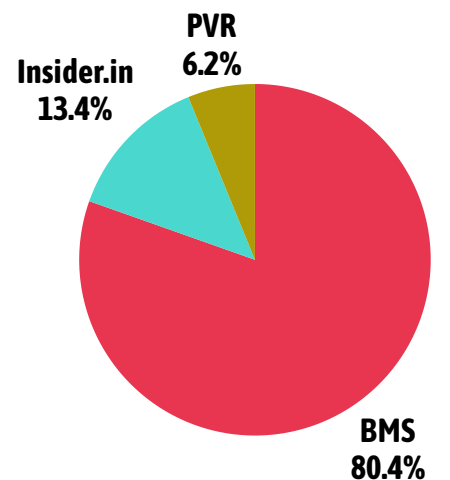


## Actors Involved



## Market Share

Source



# Problem Statement

As the **PM** of the growth team at **BookMyShow**, the goal is to resolve the **issues** users faced during **Coldplay and World Cup** bookings, so that **BMS** doesn't lose any **potential partners** in the future.

-Assumptions  
**Infinite tech bandwidth**

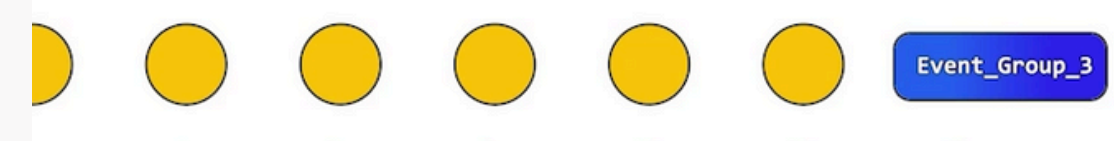
## How **BMS** is tackling the heavy traffic issue right now

- **Queue:** Virtual queue manages high demand smoothly.
- **Scaling:** AWS and Kubernetes auto-scale for traffic spikes.
- **Load Test:** "Blockbuster Days" spot issues before events.
- **Concurrency:** HAProxy limits requests to prevent overload.

## Understanding the **Virtual queue system**

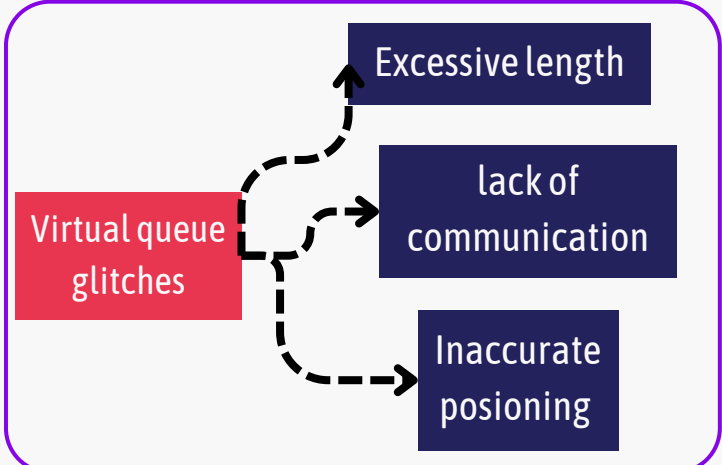
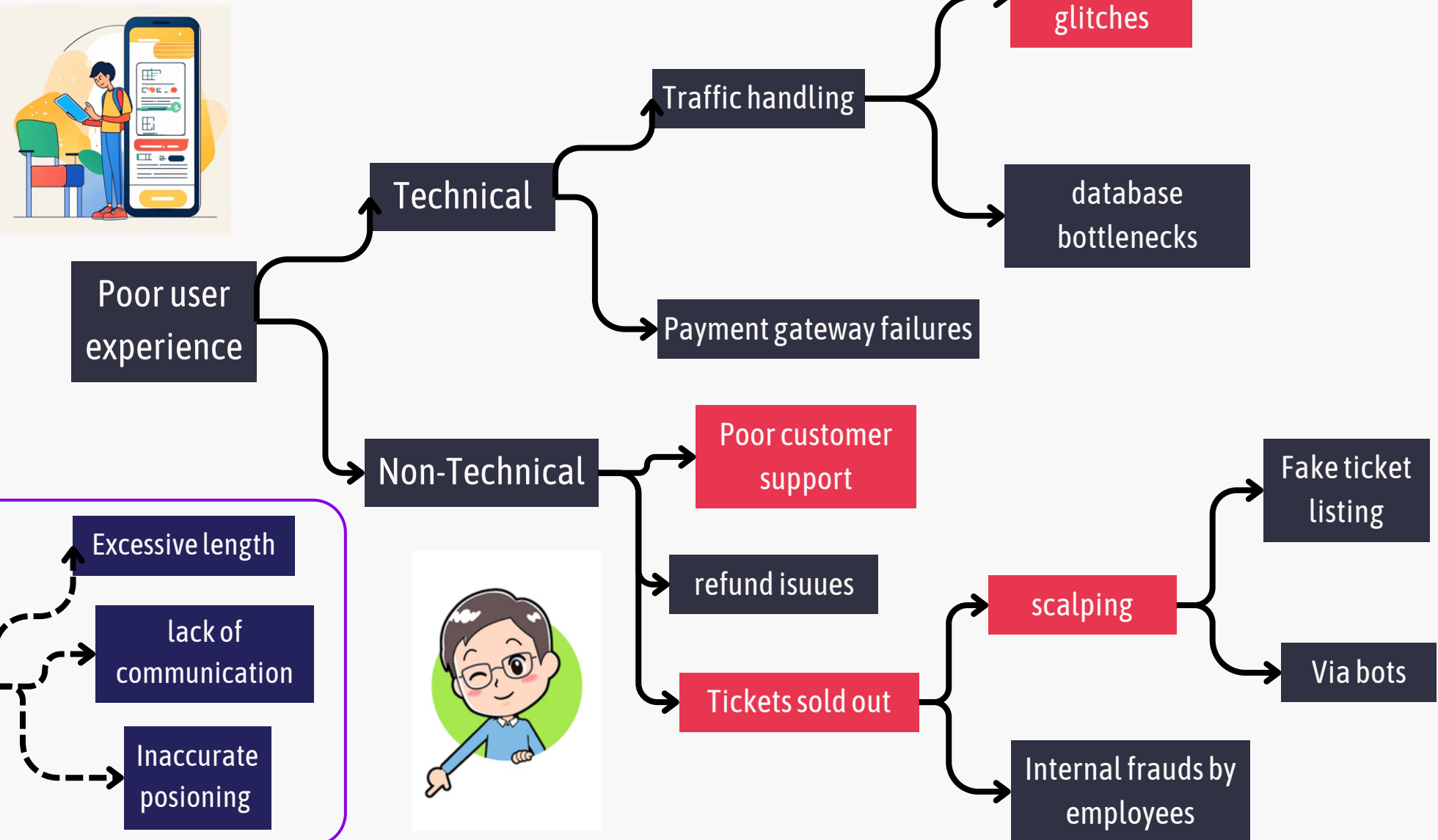
Virtual queue systems work by **automatically redirecting** visitors to a **digital queue** when they perform an action the website owner wants to protect

### Working of Virtual queue system



- This system assign each user a **position in the queue**
- The platform provide **real time update** estimated wait times and position
- This prevents server **overload** by regulating the number of concurrent requests processed.

# Mapping out Problem Outcomes



- ### What might went wrong(Hypothesis)
- **Traffic Surges:** Due to heavy traffic at the same time leads the system to collapse.
  - **Virtual Queue:** Virtual queue system was not able to solve the problem of high ticket booking properly
  - **Customer support:** Customer support was not up to the mark
  - **Scalping:** Reselling of tickets at a black price on other platforms creates frustration among users.

- ### Key outcomes
- Virtual queue glitches
  - Poor customer support
  - Scalping

- ### Source Key Trends
- Mobile booking dominance
  - Integration with streaming services
  - Rising interest of live events

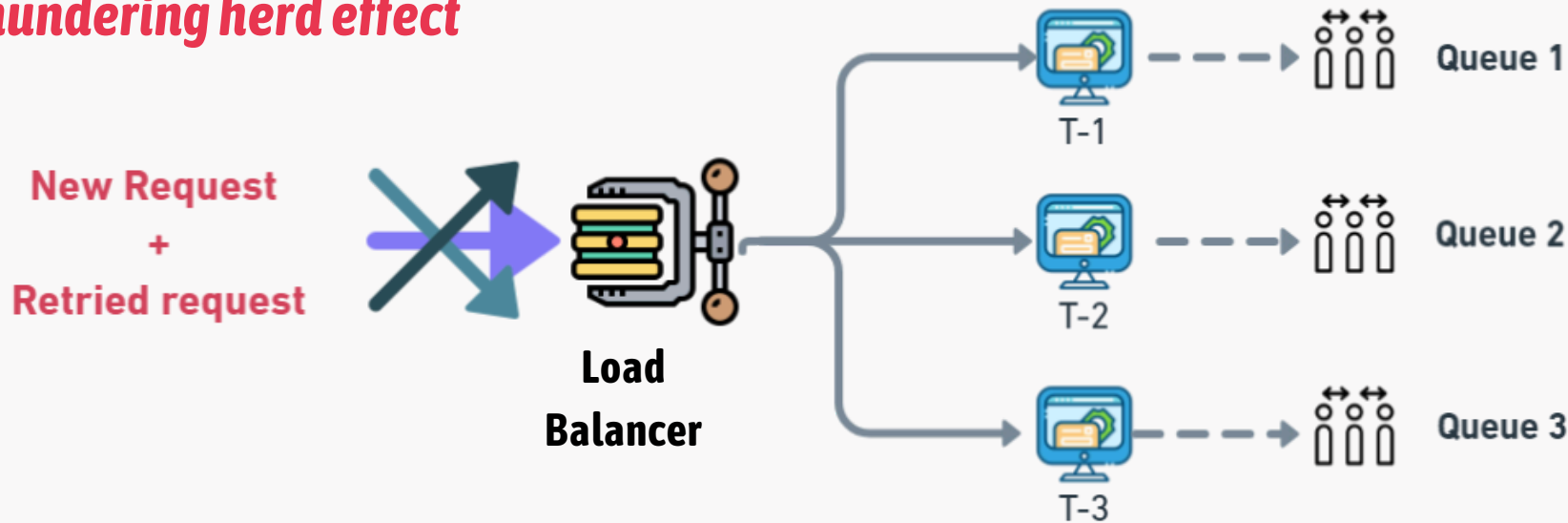
# Root cause analysis(Tech side)

To understand the root cause of what went wrong from the tech side I interview an Engineer of BMS who handles heavy traffic side



## Interview insights

### Thundering herd effect



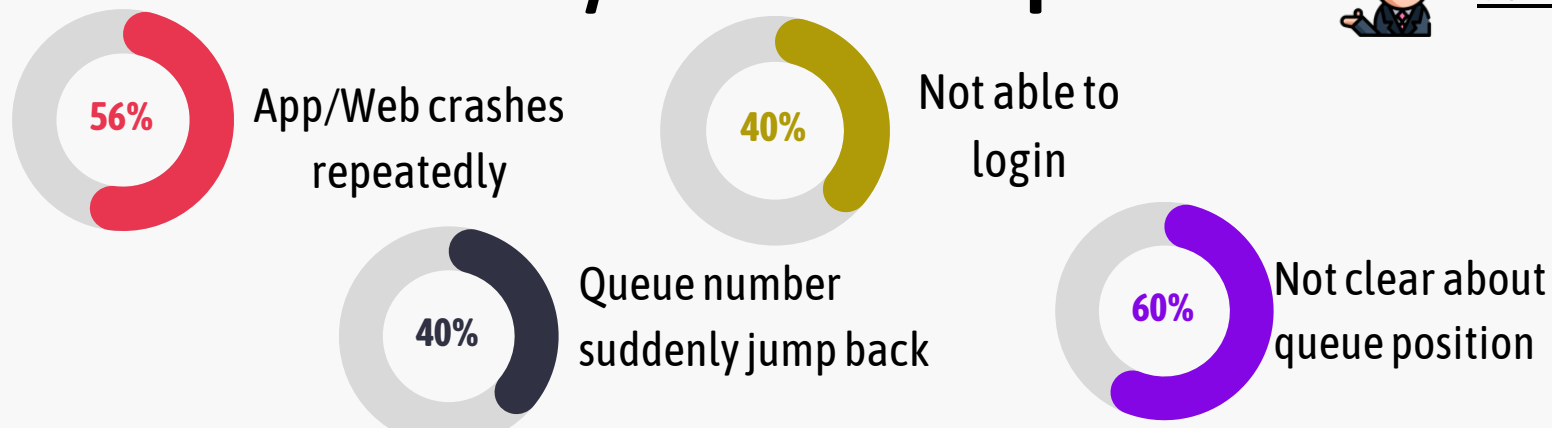
Made with Whimsical

- Load Balancers:** Spread traffic across ticket systems.
- Ticket Systems(T):** Each has its own queue for requests.
- Full Queues:** All queues fill up during heavy traffic.
- Request Denials:** New requests are denied due to no capacity
- Retried Requests:** Users retry denied requests, adding load.
- Increasing Traffic:** New and retried requests pile up.
- Cascading Effect:** Latency rises as systems reach capacity.
- Timeouts:** Latency causes requests to timeout.

## What does users say about their experience?



Survey link



## User's Insights

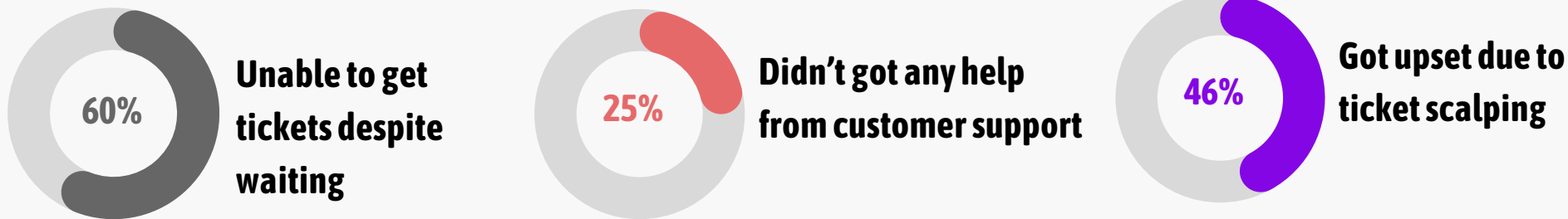


Survey link

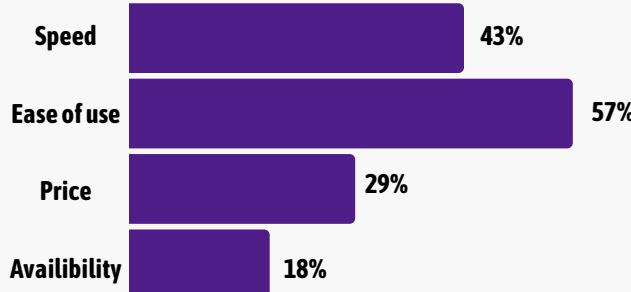
(30 responses)



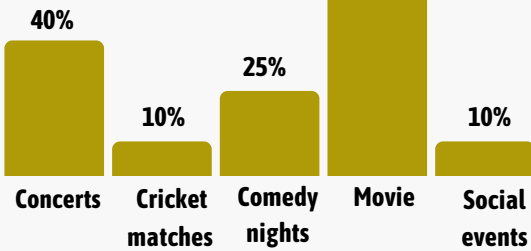
### User problems



### User Preferences

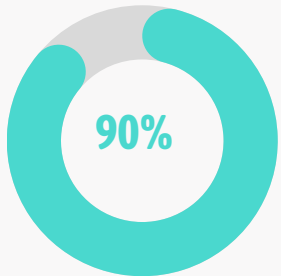


### Selection Criteria



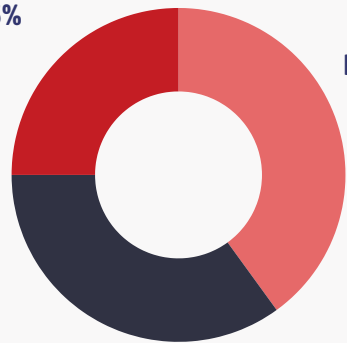
### Categories

### User Demography



Age 18-30 yrs

Priority booking  
25%



Notification in Queue  
40%

User interest

Waitlisted for Fans  
35%



### User Interviews(2 interviewee)

"I was ready 15mins prior to coldplay tickets goin in sale but the **site and app stopped** working when it was supposed to work. When th page load the **queue was in 2L**"

--Shivam,26yrs ,Bangalore

"App **keeps crashing** before the launch of any **big event**,**sells majority of big event tickets to other websites** who then sell tickets in black at **15-50X price**"

--Sakshi,22yrs ,Mumbai

### Secondary research

Source

1)**BookMyShow crashed for 15+ minutes** during Coldplay ticket sales, frustrating users with errors and long queues.

2) **13 million fans overwhelmed the system**, leaving many without tickets despite repeated attempts.

3)**Tickets priced ₹2,500-₹35,000 were resold** for up to **₹3 lakh**. BookMyShow filed complaints.

4)**Users criticized BookMyShow on social media** for failing to handle the high demand effectively.

# Problem Framing

## What is the True Problem?

As many users try to book tickets at once, the platform gets overwhelmed, causing queue delays and preventing users from securing tickets.

This negatively impacts user experience and ticket conversion rates.

## Why do we need to solve this problem now?

- 1) Guesstimate shows that BMS might lose 15% of their users if they don't solve this now
- 2) Zomato's new ticketing product might be a threat for BMS
- 3) The event ticket market in India is expected to grow at a CAGR of 2.99% from 2024 to 2028, reaching a market volume of approximately USD 2.06 billion by 2028

Source

## Value generated by solving this problem

**For users:** Faster, reliable ticket booking with less frustration and higher chances of securing tickets.

**For BMS:** Increased ticket sales, improved user satisfaction, and reduced risk of revenue loss due to system failures.

## Who are the users facing the problem?

The problem is faced by all those users who are trying to book high demand concert/Worldcup tickets and falls in the age range of 18-45 yrs

## How do we know this is the true problem?

Data from user surveys of 30 people, interviews show that system got crashed continuously and they were not able to get tickets despite waiting in the queue

## User Persona



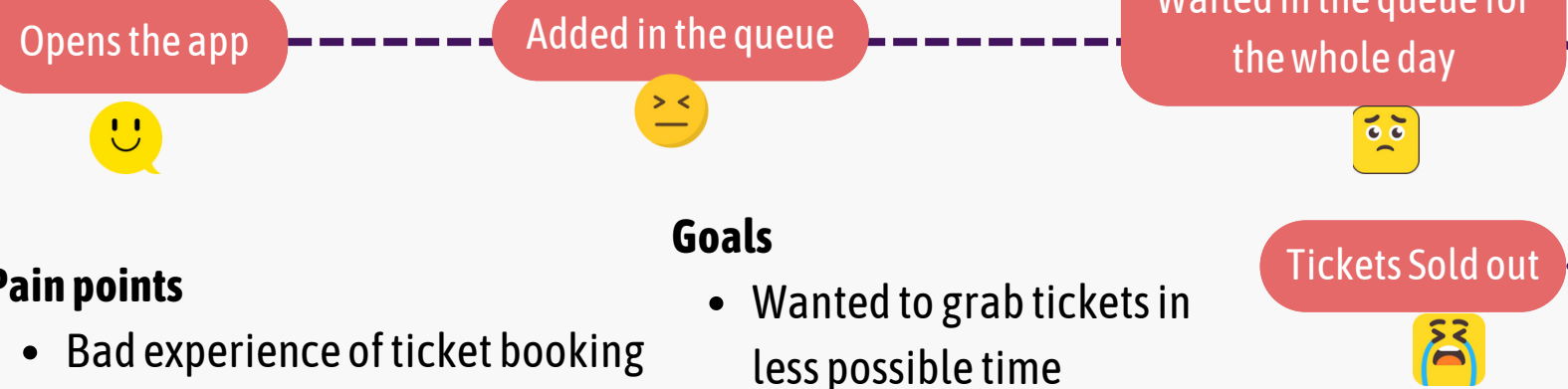
Anirudh Mishra

25 | Works in Tech | Lives in Mumbai  
Coldplay fan since 2014

"I got frustrated and not able to secure concert tickets no matter how much early I joined the sale"



### Current User Journey



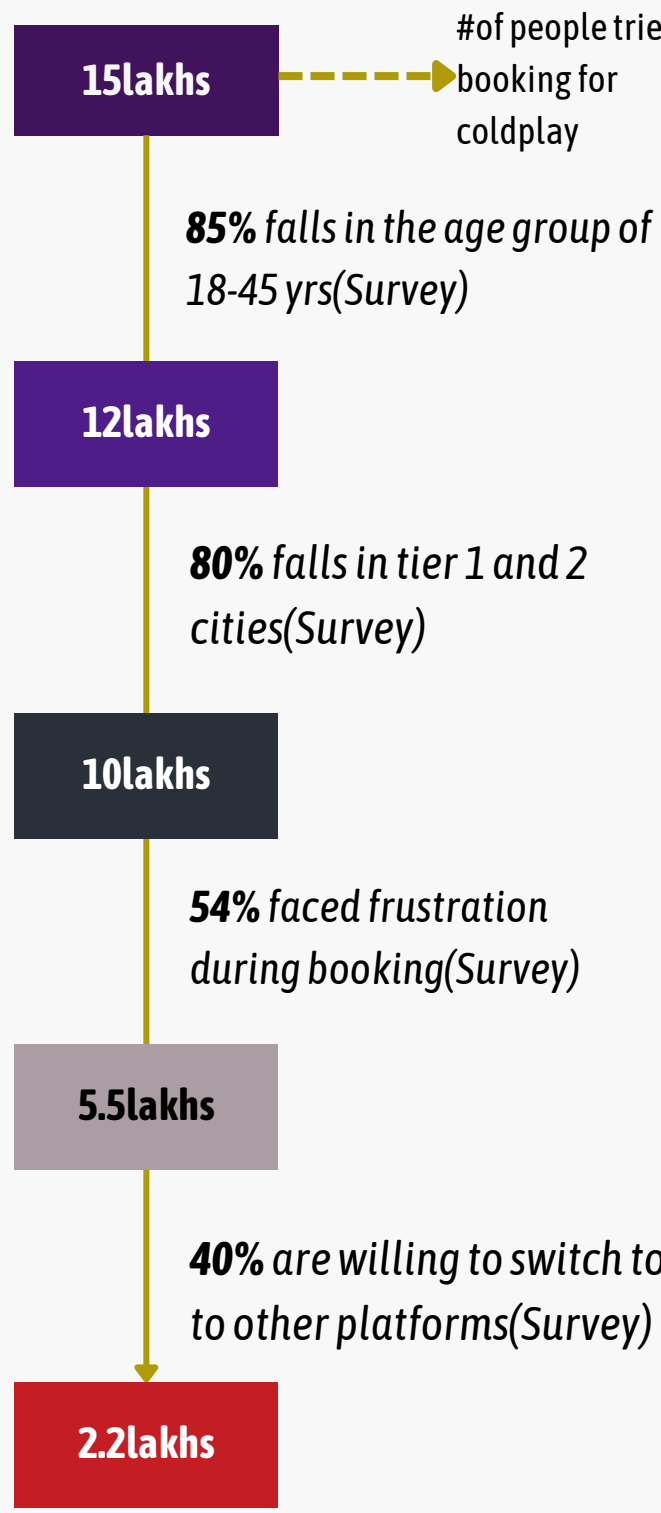
### Pain points

- Bad experience of ticket booking
- Not able to secure any ticket
- Wasted plenty of time

### Goals

- Wanted to grab tickets in less possible time
- Willing to book tickets prior to the sale

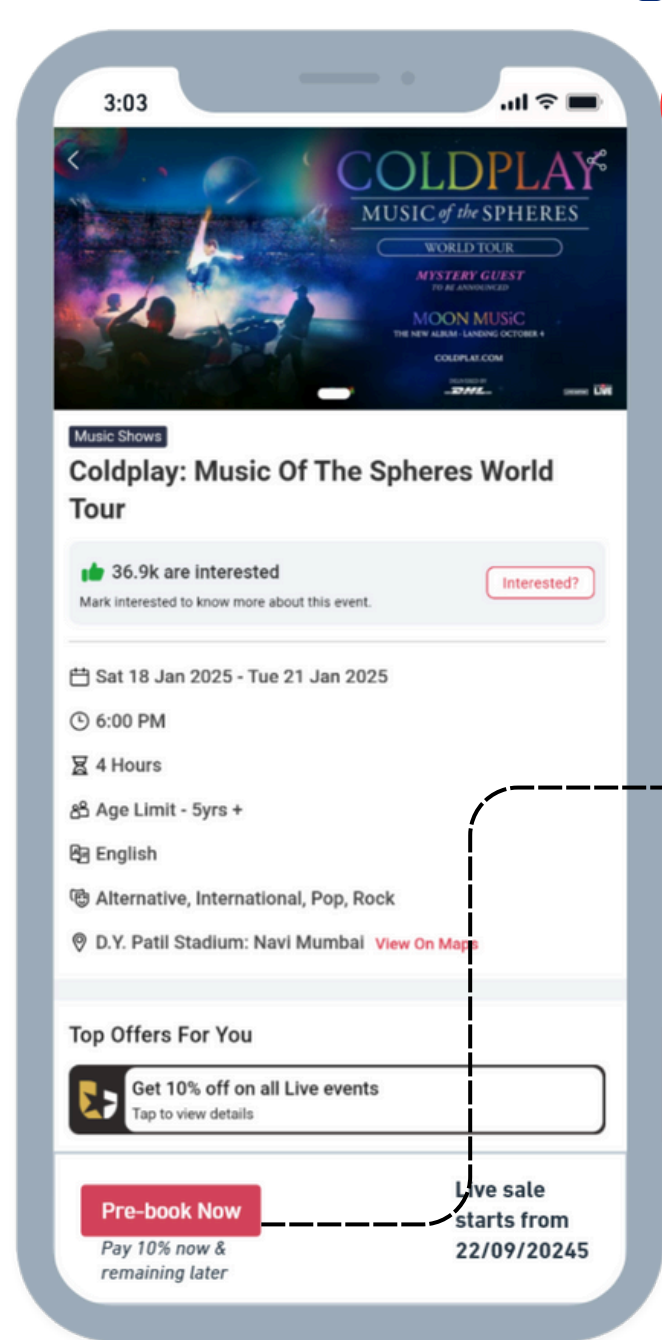
## Guesstimate on potential Impact



Out of total 15 lakhs users BMS will might lose around 2.2 lakhs (around 15%) potential users

| Proposed Solutions<br>(UX Side)                                                                                                                                                             | Demand | Effort | Adoption    | User Centricity<br>(How closely it solves the user problem)                   | Priority |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|-------------|-------------------------------------------------------------------------------|----------|
| <b>Pre- booking feature</b><br>This allows users to reserve their tickets before the official booking date by charging a small fee, easing demand during the actual sale.                   | YES    | HIGH   | Very likely | High:-Allow users to pre- book the ticket and hence reduces traffic           | V1       |
| <b>Notification System</b><br>To keep users informed about ticket availability, queue position, and other key updates during the booking process.                                           | YES    | MEDIUM | Likely      | Low:-Doesn't address the pain points of users properly                        | V2       |
| <b>Ticket Resale marketplace</b><br>This feature allows users to safely resell tickets if they are unable to attend, with other users being able to buy these tickets from the marketplace. | YES    | MEDIUM | Less likely | Medium:- This will partially address the pain points of users                 | V2       |
| <b>Scheduled ticket drops</b><br>To release tickets in small, controlled batches over a longer period to prevent server overload during a single release.                                   | YES    | MEDIUM | Likely      | High:-Allow users to book tickets in different sessions hence reduces traffic | V1       |
| <b>Waitlist feature</b><br>To give users a chance to secure tickets if the event is sold out by placing them on a waitlist for any returned or canceled tickets.                            | YES    | MEDIUM | Very likely | Medium:- This will partially address the pain points of users                 | V1       |
| <b>Interactive Waiting Room</b><br>Enhances the user experience while they wait in the virtual queue, making the waiting time more enjoyable and informative.                               | NO     | HIGH   | Less likely | Low:-Doesn't address the core pain points of users                            | V3       |
| <b>Ticket Lottery System</b><br>Tickets are randomly assigned to entrants, ensuring everyone has an equal opportunity.                                                                      | NO     | HIGH   | Less likely | Low:-users might find this unfair                                             | V3       |

# Wireframing

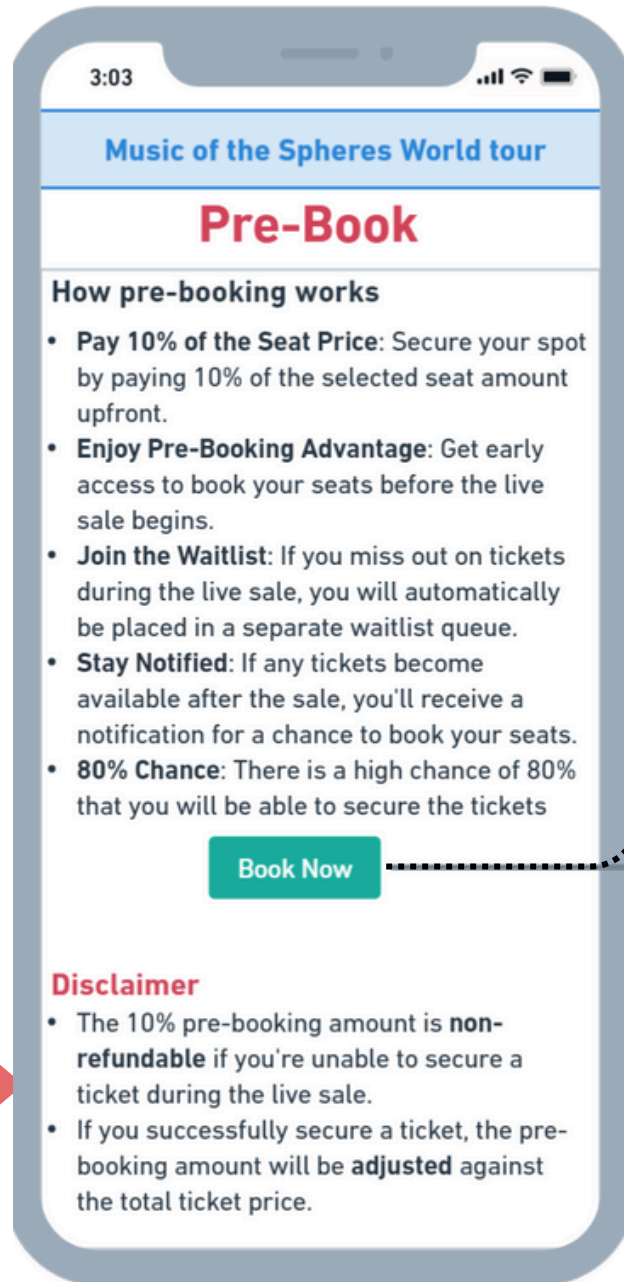


1



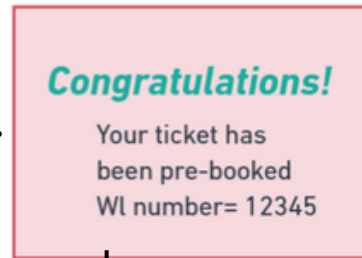
By clicking on "Pre-Book" button user can register for the event

After clicking the Pre-book button



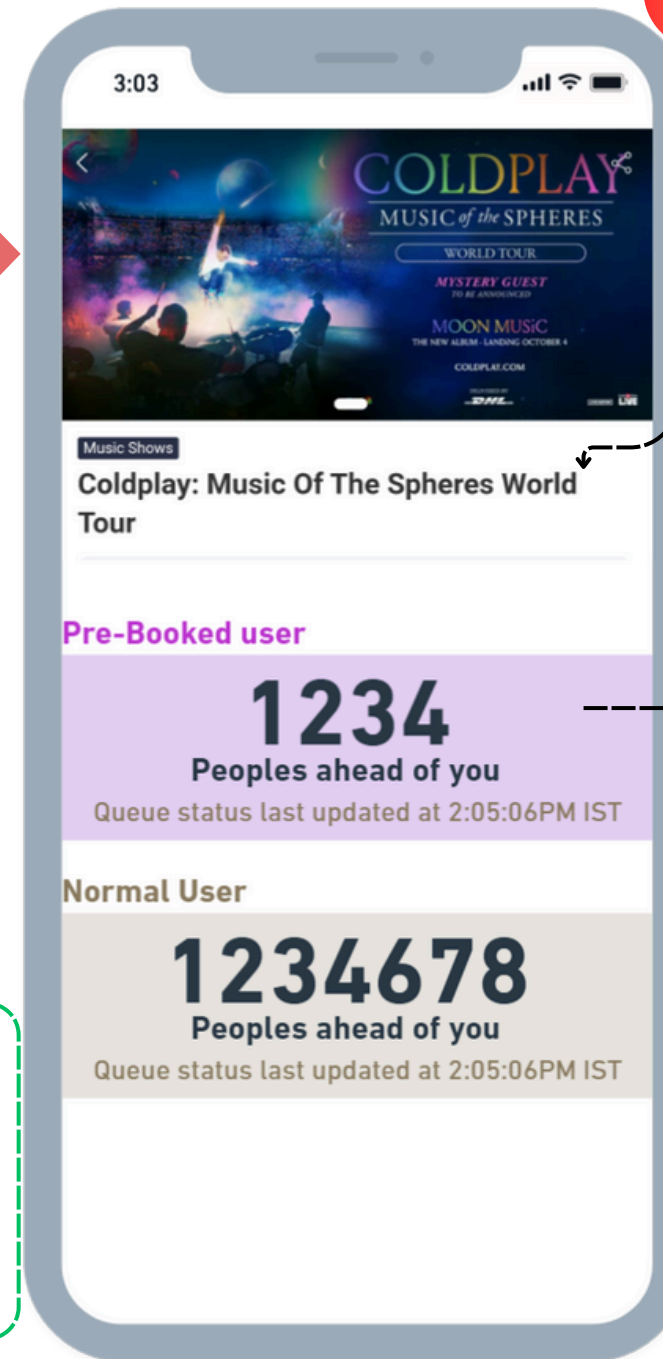
2

Once the sale event start users who have pre registered will be reredted to this page



By clicking on the "book now" button user will get this message

This message will contain a WL number which will be used during the queue



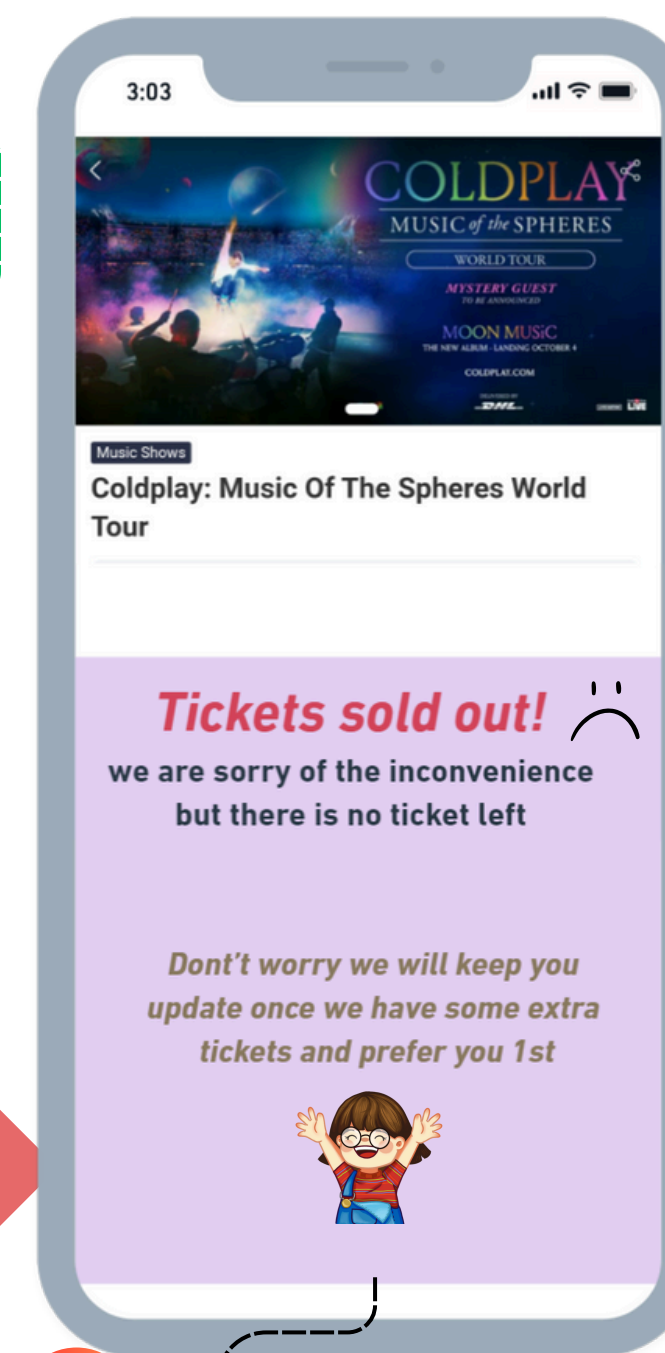
3

this screen will appear once the sale event start

this will show the number of people ahead of you based on your WL number



sale event ends and if the user not able to get a ticket



4



This screen will appear which will show the users re-availability of tickets once it is available again

## Feature Description:

- **Pay 10%** of the seat price to secure your **spot early**.
- Get access to **pre-book** before the live sale starts.
- Automatically join a **waitlist** if tickets sell out during the sale.
- Receive **notifications** if tickets become available later.
- The 10% amount is **non-refundable** if you don't get a ticket but will be **adjusted towards your ticket price** if you do.

## Value to BMS:

- Higher User Retention
- Revenue Assurance
- Enhanced User Experience
- Optimized Traffic Management:

## Value to the user:

- Increased Booking Chances
- Reduced Frustration
- Priority Access
- Convenience.



Opens the app

positive feedback

Pre-Book

Feedback loop

Wait for a short time in the queue

Got tickets easily

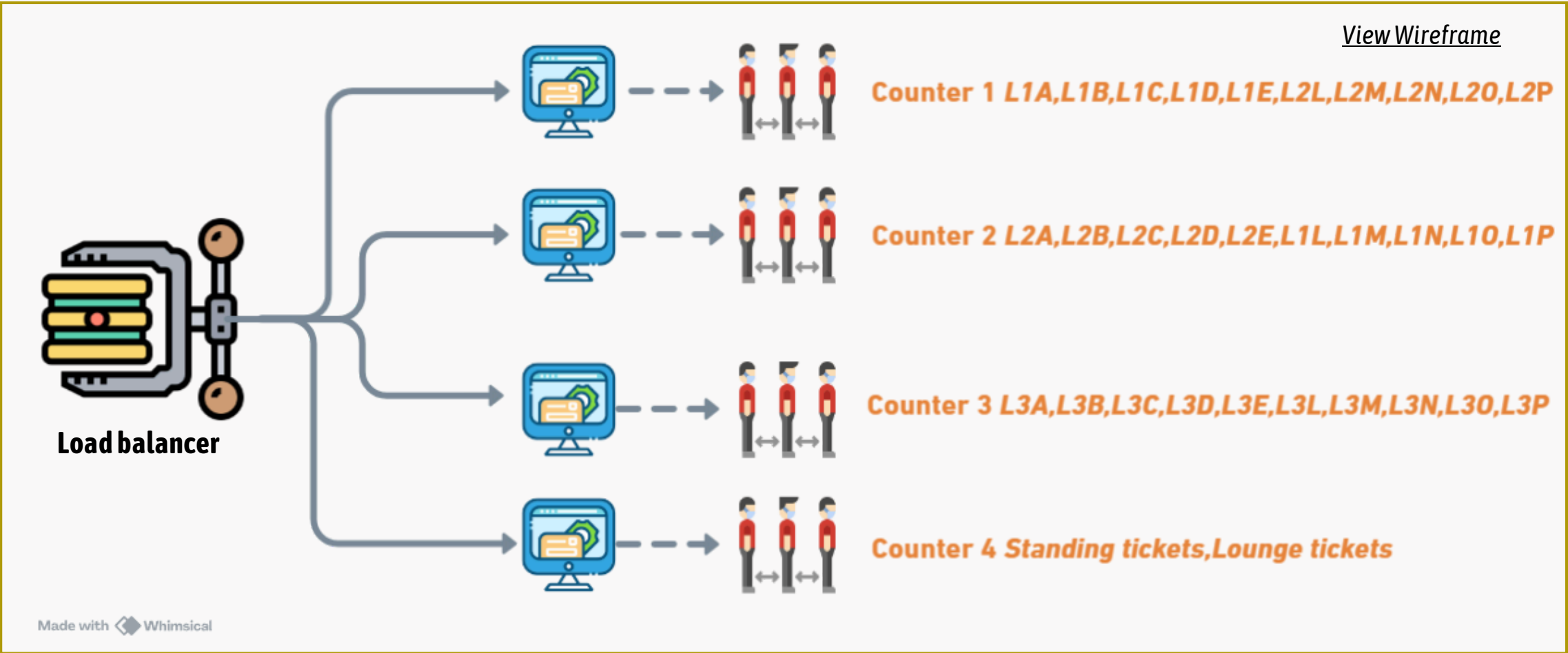
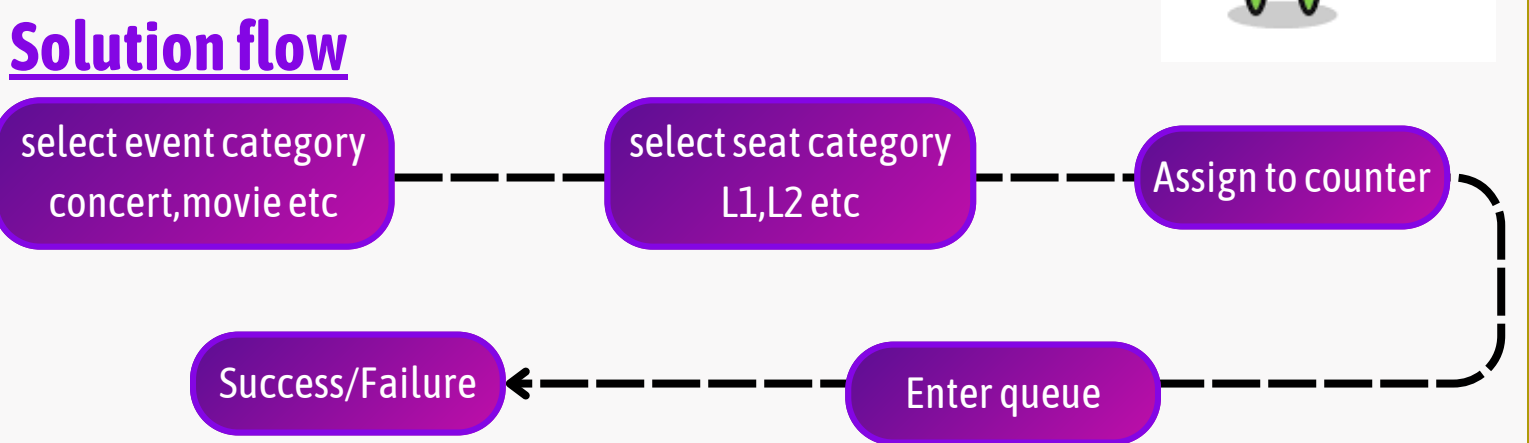
[View Wireframes](#)

| Proposed Solutions<br>(Tech side)                                                                                                                               | Demand | Effort | Adoption    | Scalability<br>(How much it can be scalable during heavy traffic)           | Priority |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|-------------|-----------------------------------------------------------------------------|----------|
| <b>Event specific queues</b><br>Grouping queues for each specific event category like concerts,cricket matches,comedy nights,movie tickets                      | YES    | MEDIUM | Likely      | Medium:- Can be scalable to a certain limit for a particular specific event | V3       |
| <b>Queues across seat categories</b><br>Implementing non overlapping ques across different different seat categories like L1 ,L2,L3,standing etc                | YES    | HIGH   | Very likely | High:-Highly scalable for a huge traffic as it is across seat category      | V1       |
| <b>Rate Limiter</b><br>Using a rate limiter can help manage high traffic during peak booking times, prevent system crashes by controlling the flow of requests. | YES    | LOW    | Likely      | High:-The device is specialized to handle huge traffic                      | V2       |

### Description of proposed solution

Split events into **non-overlapping groups**, each managed by a **separate counter or queue**. Each counter handles ticket bookings for specific **levels**, avoiding conflicts over the same ticket pool.

- Counter 1 manages:** L1A,L1B,L1C,L1D,L1E,L2L,L2M,L2N,L2O,L2P
- Counter 2 manages:** L2A,L2B,L2C,L2D,L2E,L1L,L1M,L1N,L1O,L1P
- Counter 3 manages:** L3A,L3B,L3C,L3D,L3E,L3L,L3M,L3N,L3O,L3P
- Counter 4 manages:** Standing tickets,Lounge tickets



| Important Metrics  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Metric Type                                                                                        | Metric                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Why this matters?                                                                                                                                                                      |
| NORTH STAR METRIC                                                                                  | #of users successfully completed ticket purchase/Total number of users who initiated ticket booking                                                                                                                                                                                                                                                                                                                                                                  | It tracks how well the platform handles large traffic volumes while maintaining a high conversion rate for ticket purchases.                                                           |
| ACTIVATION                                                                                         | First Pre-Book activation rate=(First time pre-booked users/Total new users)                                                                                                                                                                                                                                                                                                                                                                                         | This activation metric focuses on the adoption and initial use of the pre-booking feature, measuring how many users engage with it                                                     |
| ADOPTION                                                                                           | Pre-book usage rate=(Users who use ticket hub feature/Total number of users)                                                                                                                                                                                                                                                                                                                                                                                         | It reflects how widely the feature is being adopted by the target user base during high-demand ticket sales, indicating its popularity and usefulness.                                 |
| ENGAGEMENT                                                                                         | <ul style="list-style-type: none"> <li>#of pre-booked users who completed ticket purchase/Total pre-booked users</li> <li>Pre-booking rate:- #of user who complete the pre-book process/Total users who viewed the pre-book feature</li> <li>User interaction time:- Average time spent by users to book the final tickets</li> <li>Notification CTR:- #of users click on the notification sent to them once ticket are available/Total notification sent</li> </ul> | These metrics will help assess how users are interacting with the pre-booking feature                                                                                                  |
| RETENTION & SATISFACTION                                                                           | <ul style="list-style-type: none"> <li>Repeat pre-booking:- #of users who return to use the pre-booking feature/total users</li> <li>NPS= Net promoter score</li> <li>User retention rate:- #of users who continue using BMS after pre-book feature/total users before the feature</li> </ul>                                                                                                                                                                        | Tracking these metrics will help evaluate how well the pre-booking feature enhances user loyalty and satisfaction, while also ensuring users return to the platform for future events. |
| PERFORMANCE                                                                                        | Average ticket processing time:-The average time taken by a user to complete a ticket booking, from pre-booking to final purchase                                                                                                                                                                                                                                                                                                                                    | These metric help measure the performance of the system, ensuring that the pre-booking feature runs smoothly under high demand and enhances the overall user experience.               |

### TECH METRICS

- Response time:-The average time taken by the system to process requests
- API Request Success Rate:The percentage of successful API calls during the booking process
- Error Rate: The number of errors (such as failed bookings or system crashes) per transaction or request
- Database Query Performance: The average time taken to retrieve data from the database
- System Uptime: Percentage of time the platform is fully operational



### BUSINESS METRICS

- Incremental revenue from ticket booking:-The additional revenue generated by users who opted for pre-booking and completed their purchases.
- Increase in ticket sale per event:-The total increase in the ticket sale due after implementing this feature
- Refund and cancellation rate:-The percentage of pre-booked tickets that are canceled or refunded.



## RISKS

## Users who pre-book may fail to secure tickets during the live sale

Users may be unhappy about losing their pre-booking fee if they do not get a ticket.

Payment processing systems may fail under heavy load, causing incomplete transactions and lost bookings.

Users on the waitlist may experience delays in being notified when tickets become available

1

2

3

4

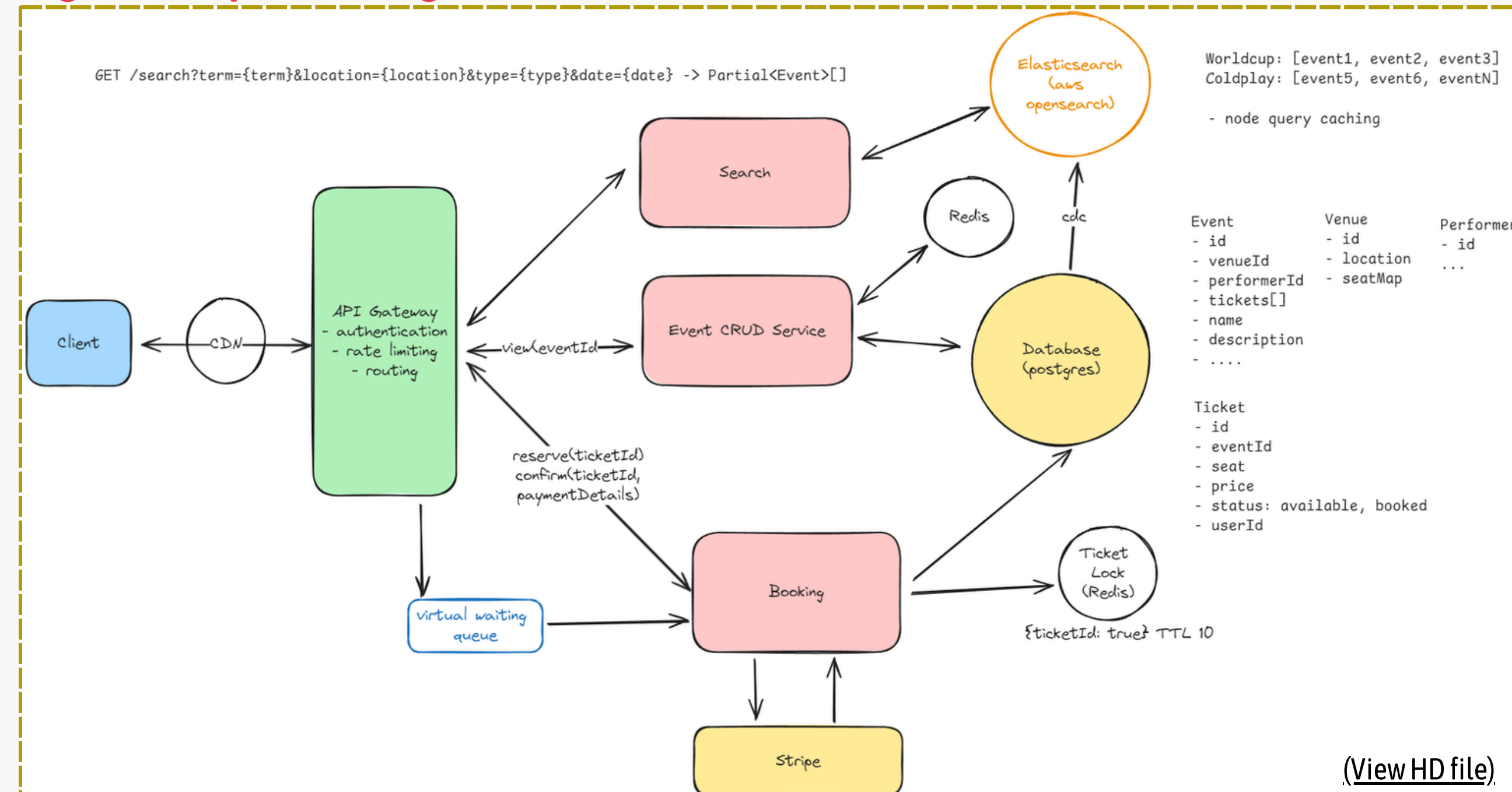
## MITIGATIONS

Offer real-time updates on queue positions and implement a transparent notification system for waitlist users.

Provide clear disclaimers upfront about the non-refundable fee

Use multiple payment gateways to handle the load and ensure backup systems are in place.

Optimize the waitlist queue management system and send real-time notifications as soon as tickets become available



## Functional Requirement

- Book tickets
- View an event
- Search for events



## Non-functional Requirements

- low latency search
- strong consistency for booking tickets & high availability for search and viewing events
- read >> write
- scalability to handle surges from popular events

(View HD file)